

Chen-Yu Kuo

郭宸毓 • Systems & full-stack engineer — earthquake early warning, from sensor firmware to the app that rings

@whes1015@gmail.comgithub.com/whes1015huggingface.co/YuYu1015

B.S. in Electrical EngineeringNational Kaohsiung Univ. of Science and Technology

Kaohsiung, Taiwan

Founder and lead engineer of **ExpTech** (探索科技), a 29-person volunteer organisation running Taiwan’s largest independent earthquake early-warning platform. I work the whole path a seismic alert travels — ESP32 firmware written register by register, the multi-region alerting fleet it reports to, the ground-motion models that decide what the alert says, and the Flutter app 100,000+ people installed. Alongside it I worked as a backend engineer at P-Waver, then as a part-time assistant at Academia Sinica’s Institute of Information Science, and now write software at the National Health Research Institutes.

Competences

Languages	TypeScript/JavaScript, Go, Python, Dart, C/C++, Rust, Lua, Shell
Systems	Linux, eBPF/XDP, Docker, Nginx/OpenResty, CoreDNS, WebSocket/SSE, Prometheus, CI/CD
Embedded	ESP32/Arduino, FreeRTOS, I ² C/SPI, LSM6DS3, GNSS (QZSS L1S), LoRa/Meshtastic
Machine Learning	PyTorch, SeisBench, ONNX, scikit-learn, vLLM, llm-compressor, TensorRT Model Optimizer
Domain	Earthquake early warning, GMPE calibration, CWA/JMA intensity scales, phase picking
Languages	Mandarin — native; English — technical reading & writing

Education

present	Dept. of Electrical Engineering, National Kaohsiung Univ. of Science & Technology
2022	B.S. in Electrical Engineering

Experience

present	Software Developer — National Health Research Institutes (國家衛生研究院), Taiwan
Jan 2025	
Jun 2025	Part-time Assistant — Institute of Information Science, Academia Sinica (中央研究院資訊科學研究所), Taipei
Dec 2024	
Nov 2024	Backend Engineer — P-Waver Inc. (衛波科技, earthquake early-warning systems), Taipei
Nov 2023	
present	Founder & Lead Engineer — ExpTech (探索科技), Taiwan
Oct 2020	- Lead a 29-member volunteer organisation across 234 repositories as one of its two administrators, across firmware, 37 Go services, Flutter clients and ML pipelines. Authored 13,411 commits inside it (19,284 across GitHub, 477 PRs) and am top contributor on every flagship codebase — DPIP 838/2,339, TREM-Lite 448/847, platform API 480/803 — and sole author of the 665-commit production infrastructure repository. - Signed a data agreement with Taiwan’s Central Weather Administration (CWA) in late 2022 — reported by 親子天下 as the Seismological Center’s first non-corporate partner; invited back 10+ times as a collaborator (天下雜誌, Feb 2026). - Ship under real stakes: 100,000+ Google Play installs and 4.2/5 from 1,378 App Store ratings for DPIP; 328,542 installer downloads across TREM desktop releases. During the M7.2 Hualien earthquake (3 Apr 2024) the network recorded 800+ events in a single day.

Machine Learning, Security & Open Source

- LLM inference & model compression — 27 models on Hugging Face (~10,600 downloads). Quantise open models to **NVFP4** (TensorRT Model Optimizer), INT4-AutoRound and GGUF for vLLM and llama.cpp, up to 35B MoE. Implemented **refusal-direction ablation** — deriving the refusal direction from paired prompts and orthogonalising it out of the weight matrices — and released the *Ornith-1.0* 9B/35B line including a DPO-tuned variant.
- Security & incident response — operated a honeypot that captured in-the-wild exploitation of **CVE-2025-55182** (React Server Components, CVSS 10.0) within days of its December 2025 disclosure; documented six malware samples with verified hashes and fifteen attacker source addresses.
- Open source — 12 merged PRs to outside projects: a rendering-performance fix to the seismograph suite **AnyShake** (spectrogram-js), a Leaflet → MapLibre migration for smc.peering.tw, and Flutter upgrades to **NKUST-AP-Flutter**, my university’s student app.
- Press — 天下雜誌, Feb 2026 (quoted as an ExpTech collaborator in a CWA Seismological Center profile); 親子天下, Jul 2024 (DPIP feature, ~500,000 downloads after the 3 Apr 2024 quake).

Jul 2026	Ground-motion & phase-picking models
Mar 2026	Python, PyTorch, SeisBench, ONNX, sole author ➤ Refit Taiwan's shaking-intensity model on 45,398 CWA PGA/PGV/intensity observations (2000–2025), replacing a legacy equation that over-amplified with magnitude. Added a direct PGV prediction equation and corrected a site-amplification double-count that inflated estimates $\sim 2\times$. Held-out: 62.0% exact intensity vs 39.7% for the model then in production, MAE 0.400 vs 0.671, 0% false alarms. ➤ Trained a 50 Hz PhaseNet P/S phase picker (6,457-line pipeline: dataset ingest, noise injection, evaluation, ONNX export). Diagnosed a windowing bug where the source window was shorter than the model window, teaching the model that P always sits mid-window and emitting a false pick every 45 s stride. Both models published on Hugging Face.
Jun 2026	ES-Net — seismometer firmware & national sensor network
Sep 2025	C++, ESP32, FreeRTOS, sole author, 7,759 LOC ➤ Hand-wrote an LSM6DS3 accelerometer driver at register level — own register map, who_am_i, ctrl1_xl, int1_ctrl and FIFO paths — rather than depending on a vendor library. ➤ On-device JMA seismic-intensity computation: an IIR filter bank plus a binary indexed tree for the running amplitude percentile the JMA scale requires. ➤ Timing is the hard problem here: SNTP in smooth mode so the clock slews rather than steps mid-waveform, plus HW/SW watchdogs, OTA update, and a documented custom binary wire protocol with TOTP station authentication. ➤ Feeds 200+ TREM-Net stations nationwide; first-generation units were self-assembled for under NT\$800 each.
present	TREM — real-time earthquake monitoring desktop platform
Jan 2023	JavaScript, Electron, Rust, top contributor, 448/847 commits ➤ Desktop client rendering live ground motion from TREM-Net and CWA feeds; 93 GitHub stars and 328,542 installer downloads across 74 releases, shipped continuously since Jan 2023. Designed the plugin system third parties extend it through and maintain its registry (417 commits); wrote the trem-monitor station-health service in Rust (105/150).
present	Production infrastructure
Dec 2025	Go, eBPF/XDP, OpenResty, CoreDNS, sole author ➤ kekikai — packet filter in Go using cilium/ebpf, with an XDP ingress program and a TC egress hook; lru_hash/array/ringbuf maps drive per-IP rate limiting, allow/blocklists and port policy, with an emergency detach path. ➤ Nginx/OpenResty fleet — 23 upstreams across nodes in Taipei, Kaohsiung, Tainan, Tokyo and Osaka, with Lua modules for traffic accounting, NTP and image proxying. ➤ Split-horizon CoreDNS with a custom Go dnsstats plugin (miekg/dns) exposing per-domain/client query stats and Prometheus metrics. Sole author too of the NTP server, API gateway and Meshtastic LoRa ingest service.
present	DPiP — cross-platform disaster alerting app
Aug 2023	Dart/Flutter, TypeScript, top contributor, 838/2,339 commits ➤ iOS and Android client integrating TREM-Net and CWA feeds with push alerting, MapLibre layers and 10 interface languages. Wrote the test suite and custom commit-gate tooling outright, and authored the backend service dpip-server-ts in full.
present	QZSS L1S disaster-message decoder
Aug 2026	C++, sole author, $\sim 12k$ LOC with unit tests ➤ Decodes QZSS L1S 250-bit SBAS frames end to end — preamble, 6-bit message type, 212-bit payload and CRC-24Q — including type 43 DC Report (JMA disaster broadcasts, IS-QZSS-DCR) and type 44 DCX.